



ASSIGNMENT - 02

EDA with Python

DATA SCIENCE

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EXPLORATORY DATA ANALYSIS USING PYTHON

1. What is Exploratory Data Analysis (EDA) and why is it important?
2. Which Python libraries are commonly used for EDA and data manipulation?
3. How do you read a CSV file into a Pandas DataFrame?
4. How would you display the first few rows of a DataFrame?
5. How can you check the dimensions (number of rows and columns) of a DataFrame?
6. What does the describe() method provide when called on a DataFrame?
7. How do you handle missing values in a dataset? Name two methods.
8. How can you detect outliers in a dataset?
9. What is the difference between the .loc and .iloc indexers in Pandas?
10. How do you filter rows in a DataFrame based on a condition on a column?
11. How do you add a new column to an existing Pandas DataFrame?
12. What is the Pandas groupby function used for? Give an example.
13. How would you merge or join two DataFrames in Pandas?
14. What is the difference between an inner join and a left join when merging DataFrames?
15. How can you remove duplicate rows from a DataFrame?
16. What is a pivot table in Pandas and when might you use one?
17. How do you reshape data from wide format to long format (or vice versa) in Pandas?
18. How can you calculate the correlation between all pairs of numerical columns in a DataFrame?
19. How do you create a histogram of a column in a dataset using Matplotlib or Pandas?
20. Which type of plot would you use to visualize the relationship between two numerical variables?
21. What is a box plot and what can it tell you about a dataset?
22. How can you identify which columns have missing values and how many missing values they have?
23. How do you work with date or time data in Pandas (e.g., parsing dates or resampling time series data)?
24. How would you sort a DataFrame by the values of one or more columns?
25. What is one-hot encoding and how can you perform it using Pandas?
26. How can you combine multiple DataFrames vertically (stacking one on top of another)?
27. How do you randomly sample a fraction of rows from a DataFrame?
28. What is a heatmap and how might it be useful during EDA?
29. How do you save or export a Pandas DataFrame to a CSV file?
30. What are some common data quality issues you would check for during EDA?